

INTEGRALES TRIGONOMÉTRICAS

Propósito: Determinar integrales trigonométricas indefinidas.

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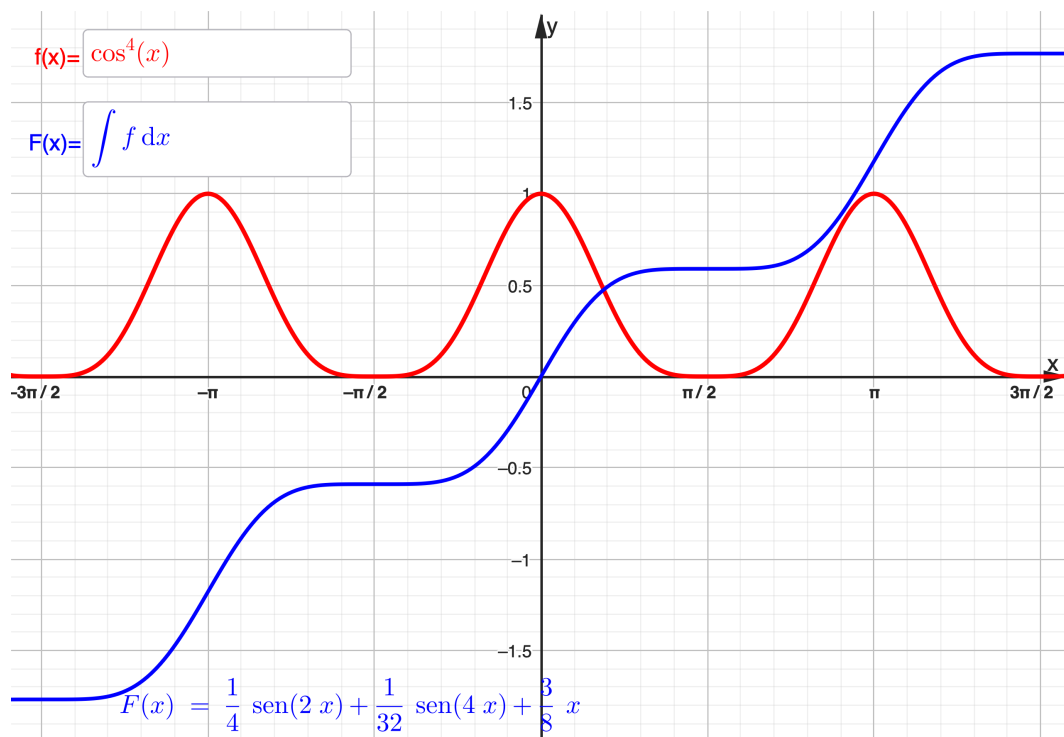
Identidades trigonométricas a usar:

Identidad Pitagórica	Identidad de reducción de potencia para $\sin^2(x)$	Identidad de reducción de potencia para $\cos^2(x)$
$\sin^2(x) + \cos^2(x) = 1$	$\sin^2(x) = \frac{1 - \cos(2x)}{2}$	$\cos^2(x) = \frac{1 + \cos(2x)}{2}$

Ejemplo: Evaluar

$$\begin{aligned}
 & \int \cos^4(x) \, dx \\
 &= \int [\cos^2(x)]^2 \, dx \\
 &= \int \left(\frac{1 + \cos(2x)}{2} \right)^2 \, dx \\
 &= \int \frac{[1 + \cos(2x)]^2}{2^2} \, dx \\
 &= \frac{1}{4} \int [1 + 2 \cos(2x) + \cos^2(2x)] \, dx \\
 &= \frac{1}{4} \left[\int dx + 2 \int \cos(2x) \, dx + \int \cos^2(2x) \, dx \right] \\
 &= \frac{1}{4} \left[\int dx + 2 \int \cos(2x) \, dx + \int \left(\frac{1 + \cos(4x)}{2} \right) \, dx \right] \\
 &= \frac{1}{4} \left[x + 2 \int \cos(2x) \, dx + \int \left(\frac{1 + \cos(4x)}{2} \right) \, dx \right] \\
 &= \frac{1}{4} \left[x + \frac{2 \sin(2x)}{2} + \int \left(\frac{1 + \cos(4x)}{2} \right) \, dx \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{4} \left[x + \sin(2x) + \frac{1}{2} \int (1 + \cos(4x)) dx \right] \\
 &= \frac{1}{4} \left[x + \sin(2x) + \frac{1}{2} \left(\int dx + \int \cos(4x) dx \right) \right] \\
 &= \frac{1}{4} \left[x + \sin(2x) + \frac{1}{2} \left(x + \frac{\sin(4x)}{4} \right) \right] + C \\
 &= \frac{1}{4} \left[x + \sin(2x) + \frac{x}{2} + \frac{\sin(4x)}{8} \right] + C \\
 &= \frac{x}{4} + \frac{\sin(2x)}{4} + \frac{x}{8} + \frac{\sin(4x)}{32} + C \\
 &= \frac{3x}{8} + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C
 \end{aligned}$$



Ejemplo: Evaluar

$$\int \sin^5(x) \cos^2(x) \, dx$$

$$= \int \sin^4(x) \cos^2(x) \sin(x) dx$$

$$\int [\sin^2(x)]^2 \cos^2(x) \sin(x) dx$$

$$\int [1 - \cos^2(x)]^2 \cos^2(x) \sin(x) dx$$

$$u = \cos(x), \quad du = -\sin(x) dx$$

$$= - \int [1 - u^2]^2 u^2 du$$

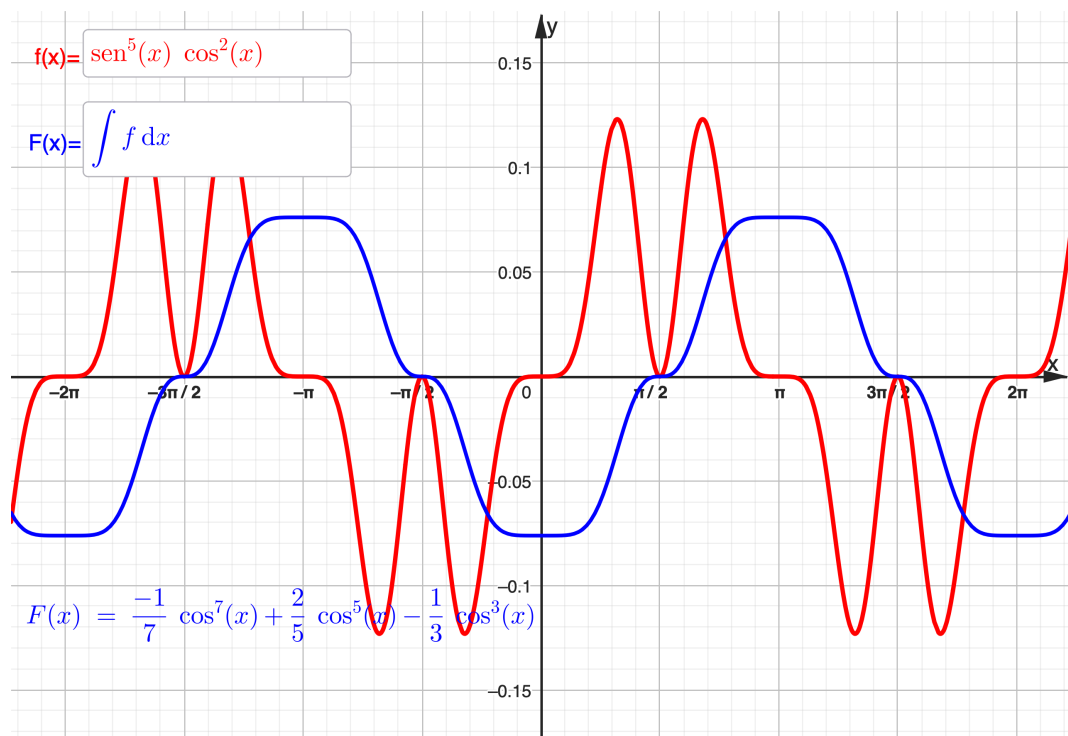
$$= - \int [1 - 2u^2 + u^4] u^2 du$$

$$= - \int [u^2 - 2u^4 + u^6] du$$

$$= - \int u^2 du + 2 \int u^4 du - \int u^6 du$$

$$= -\frac{u^3}{3} + \frac{2u^5}{5} - \frac{u^7}{7} + C$$

$$= -\frac{\cos^3(x)}{3} + \frac{2\cos^5(x)}{5} - \frac{\cos^7(x)}{7} + C$$



ESTRATEGIA PARA LA EVALUACIÓN DE

$$\int \sin^m(x) \cos^n(x) dx$$

- a) Si la potencia del coseno es impar ($n = 2k + 1$), extraemos un factor coseno y utilizamos $\cos^2(x) = 1 - \sin^2(x)$ para expresar los factores restantes en términos del seno:

$$\begin{aligned} \int \sin^m(x) \cos^{2k+1}(x) dx &= \int \sin^m(x) (\cos^2(x))^k \cos(x) dx \\ &= \int \sin^m(x) (1 - \sin^2(x))^k \cos(x) dx \end{aligned}$$

Después sustituimos $u = \sin(x)$, obteniéndose $du = \cos(x) dx$.

- b) Si la potencia del seno es impar ($m = 2k + 1$), extraemos un factor seno y utilizamos $\sin^2(x) = 1 - \cos^2(x)$ para expresar los factores restantes en términos del coseno:

$$\begin{aligned} \int \sin^{2k+1}(x) \cos^n(x) dx &= \int (\sin^2(x))^k \cos^n(x) \sin(x) dx \\ &= \int (1 - \cos^2(x))^k \cos^n(x) \sin(x) dx \end{aligned}$$

Después sustituimos $u = \cos(x)$, obteniéndose $du = -\sin(x) dx$

Observación: Si la potencia de ambos, seno y coseno, es impar, puede utilizarse a) o b).

- c) Si la potencia de ambos, seno y coseno, son pares, utilizamos las identidades de reducción de potencias o del ángulo medio.

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}, \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

Algunas veces es útil utilizar la identidad

$$\sin(x) \cos(x) = \frac{\sin(2x)}{2}$$

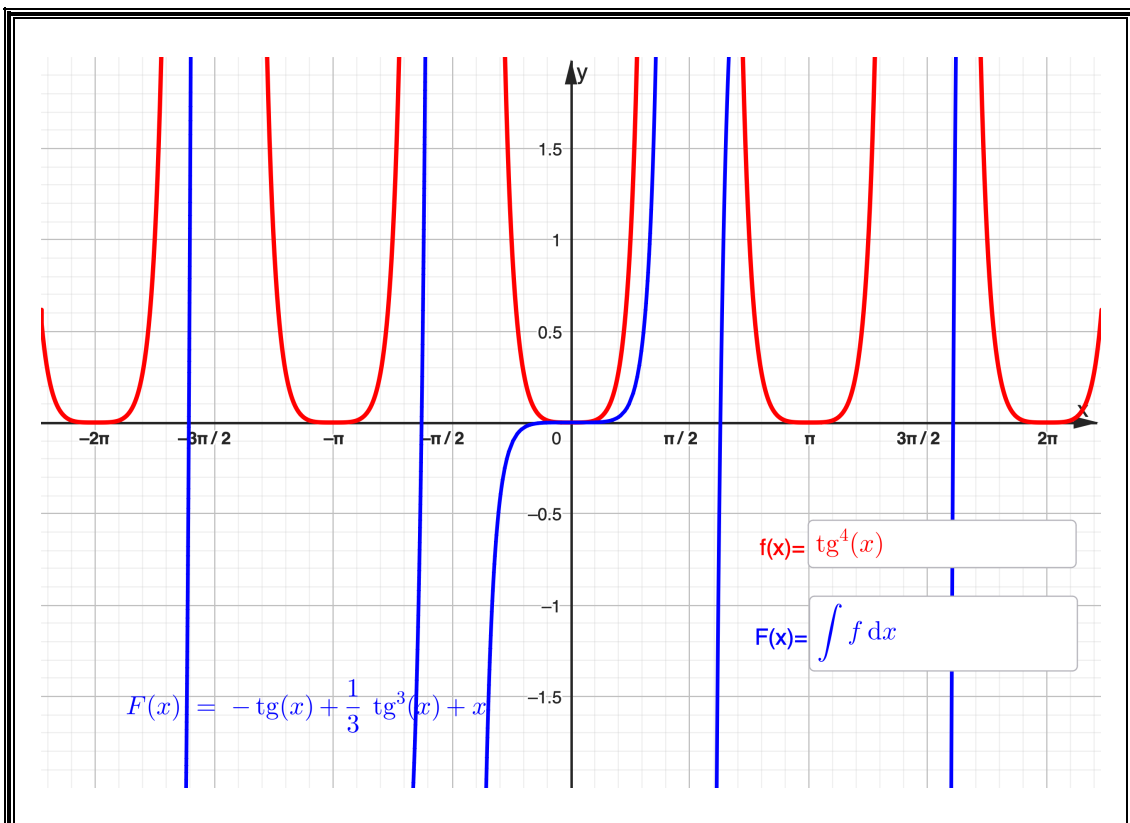
$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

Identidades trigonométricas a usar:

Identidades Trigonómicas	
$\tan^2(x) + 1 = \sec^2(x)$	$\tan^2(x) = \sec^2(x) - 1$

Ejemplo: Evaluar

$$\begin{aligned} & \int \tan^4(x) \, dx \\ &= \int \tan^2(x) \tan^2(x) \, dx \\ &= \int \tan^2(x) [\sec^2(x) - 1] \, dx \\ &= \int [\tan^2(x) \sec^2(x) - \tan^2(x)] \, dx \\ &= \int \tan^2(x) \sec^2(x) \, dx - \int \tan^2(x) \, dx \\ &= \int \tan^2(x) \sec^2(x) \, dx - \int [\sec^2(x) - 1] \, dx \\ &= \int \tan^2(x) \sec^2(x) \, dx - \int \sec^2(x) \, dx + \int 1 \, dx \\ & \quad u = \tan(x), \quad du = \sec^2(x) \, dx \\ &= \int u^2 \, du - \int \sec^2(x) \, dx + \int 1 \, dx \\ &= \frac{u^3}{3} - \int \sec^2(x) \, dx + \int 1 \, dx \\ &= \frac{\tan^3(x)}{3} - \int \sec^2(x) \, dx + \int 1 \, dx \\ &= \frac{\tan^3(x)}{3} - \tan(x) + x + C \end{aligned}$$



Ejemplo: Evaluar

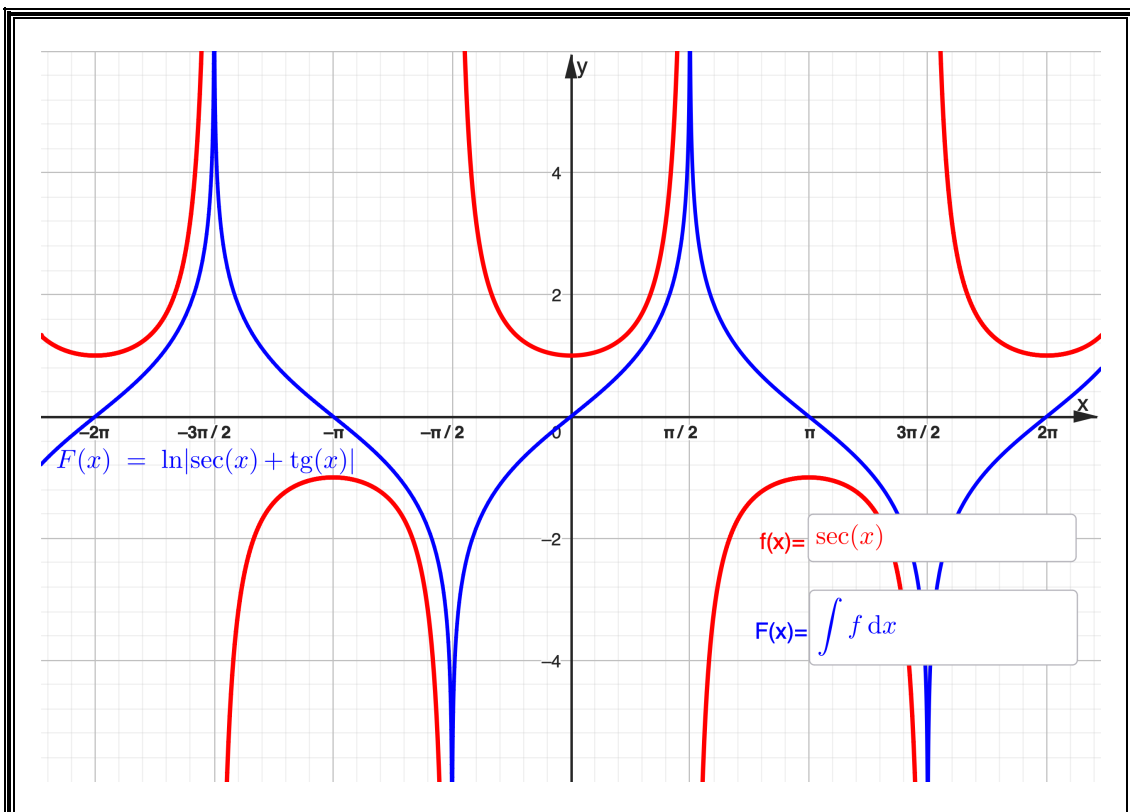
$$\begin{aligned}
 & \int \sec(x) \, dx \\
 &= \int \sec(x) \left(\frac{\sec(x) + \tan(x)}{\sec(x) + \tan(x)} \right) dx \\
 &= \int \frac{\sec^2(x) + \sec(x) \tan(x)}{\sec(x) + \tan(x)} dx
 \end{aligned}$$

$$u = \sec(x) + \tan(x), \quad du = \sec(x) \tan(x) + \sec^2(x) \, dx$$

$$= \int \frac{1}{u} du$$

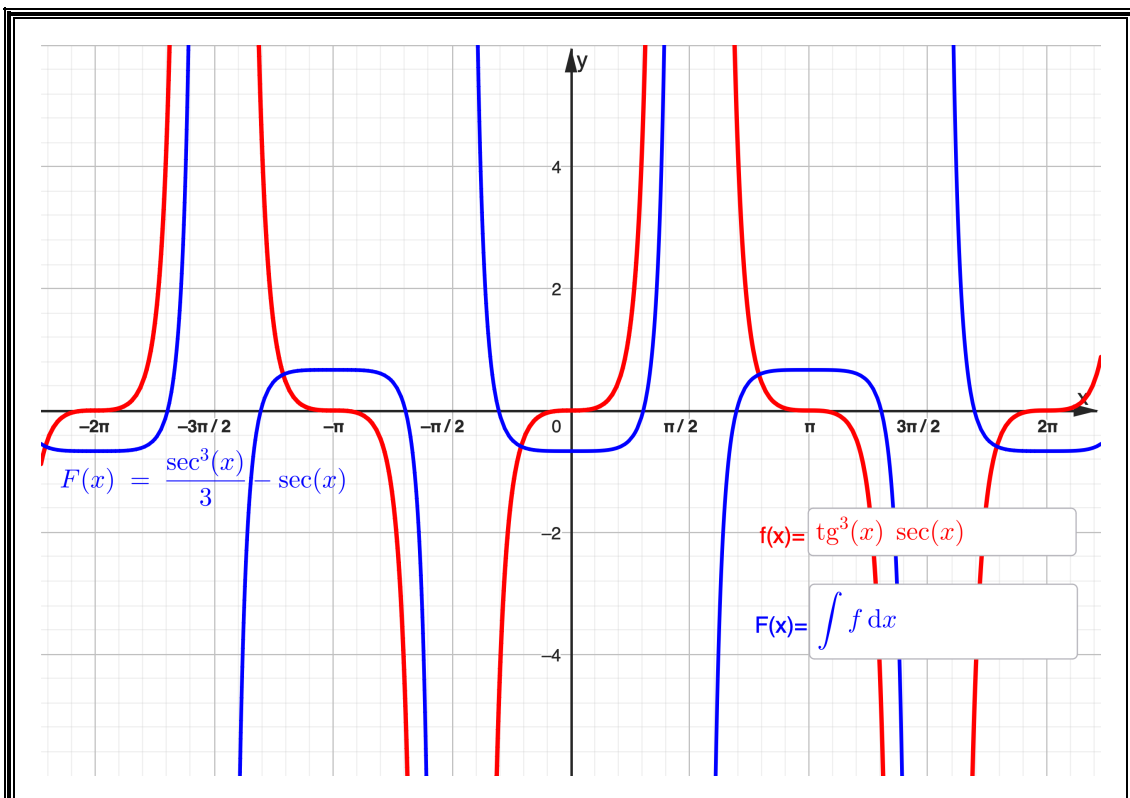
$$= \ln|u| + C$$

$$= \ln|\sec(x) + \tan(x)| + C$$



Ejemplo: Evaluar

$$\begin{aligned}
 & \int \tan^3(x) \sec(x) dx \\
 &= \int \tan^2(x) \tan(x) \sec(x) dx \\
 &= \int (\sec^2(x) - 1) \tan(x) \sec(x) dx \\
 &= \int [\sec^2(x) \tan(x) \sec(x) - \tan(x) \sec(x)] dx \\
 &= \int \sec^2(x) \tan(x) \sec(x) dx - \int \tan(x) \sec(x) dx \\
 & \quad u = \sec(x), \quad du = \sec(x) \tan(x) dx \\
 &= \int u^2 du - \int \tan(x) \sec(x) dx \\
 &= \frac{u^3}{3} - \sec(x) + C \\
 &= \frac{\sec^3(x)}{3} - \sec(x) + C
 \end{aligned}$$



ESTRATEGIA PARA LA EVALUACIÓN DE

$$\int \tan^m(x) \sec^n(x) dx$$

- a) Si la potencia de la secante es par ($n = 2k, k \geq 2$), extraemos un factor $\sec^2(x)$ y utilizamos $\sec^2(x) = 1 + \tan^2(x)$ para expresar los factores restantes en términos de $\tan(x)$:

$$\begin{aligned} \int \tan^m(x) \sec^{2k}(x) dx &= \int \tan^m(x) (\sec^2(x))^{k-1} \sec^2(x) dx \\ &= \int \tan^m(x) (1 + \tan^2(x))^{k-1} \sec^2(x) dx \end{aligned}$$

Después sustituimos $u = \tan(x)$, obteniéndose $du = \sec^2(x) dx$

- b) Si la potencia de la tangente es impar ($m = 2k + 1$), extraemos un factor $\sec(x) \tan(x)$ y utilizamos $\tan^2(x) = \sec^2(x) - 1$ para expresar los factores restantes en términos de $\sec(x)$:

$$\begin{aligned} \int \tan^{2k+1}(x) \sec^n(x) dx &= \int (\tan^2(x))^k \sec^{n-1}(x) \sec(x) \tan(x) dx \\ &= \int (\sec^2(x) - 1)^k \sec^{n-1}(x) \sec(x) \tan(x) dx \end{aligned}$$

Después sustituimos $u = \sec(x)$, obteniéndose $du = \sec(x) \tan(x) dx$.

Para evaluar las integrales

$$\text{a)} \int \sin(mx) \cos(nx) dx$$

$$\text{b)} \int \sin(mx) \sin(nx) dx$$

$$\text{c)} \int \cos(mx) \cos(nx) dx$$

utilizamos la identidad correspondiente

a)	$\sin(A) \cos(B) = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$
b)	$\sin(A) \sin(B) = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$
c)	$\cos(A) \cos(B) = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$

Ejemplo: Evaluar

$$\int \sin(8x) \cos(5x) dx$$

$$\sin(A) \cos(B) = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$$

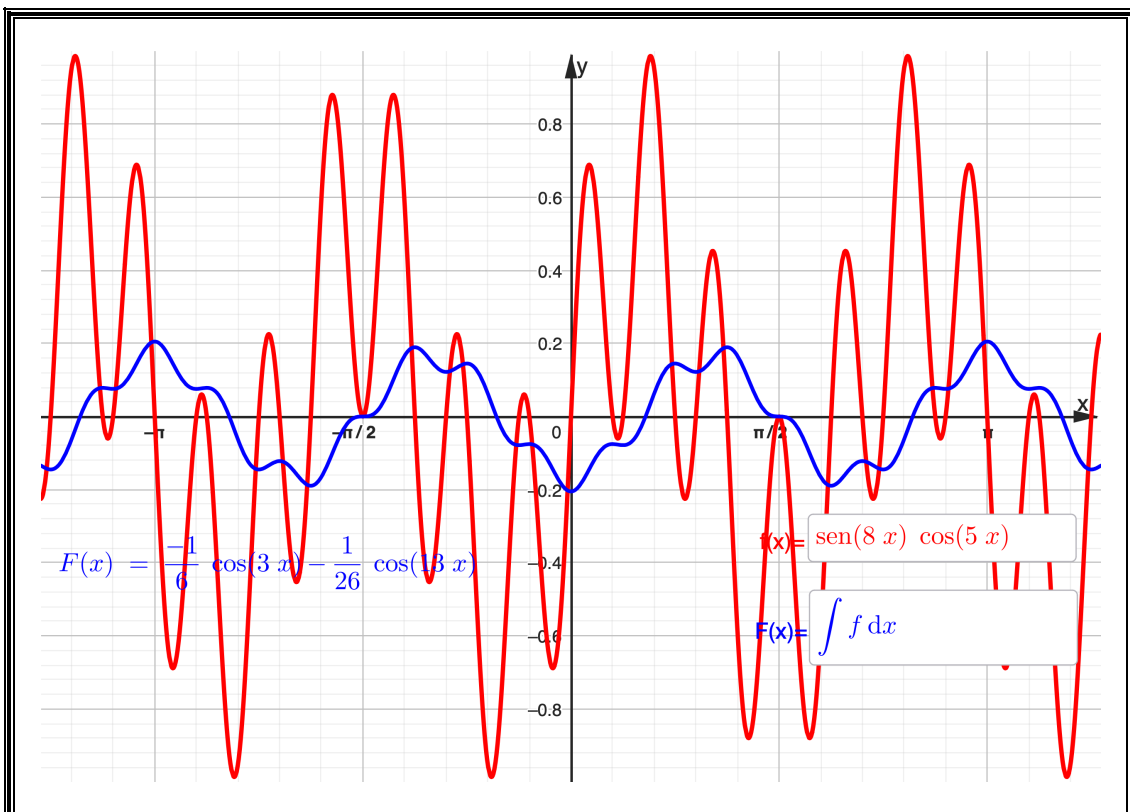
$$= \int \frac{1}{2} [\sin(8x - 5x) + \sin(8x + 5x)] dx$$

$$= \frac{1}{2} \int [\sin(3x) + \sin(13x)] dx$$

$$= \frac{1}{2} \left[\int \sin(3x) dx + \int \sin(13x) dx \right]$$

$$= \frac{1}{2} \left[-\frac{1}{3} \cos(3x) - \frac{1}{13} \cos(13x) + C \right]$$

$$= -\frac{1}{6} \cos(3x) - \frac{1}{26} \cos(13x) + C$$



Ejemplo: Evaluar

$$\int \frac{\sin(\beta)}{\cos^3(\beta)} d\beta$$

$$u = \cos(\beta), \quad du = -\sin(\beta) d\beta$$

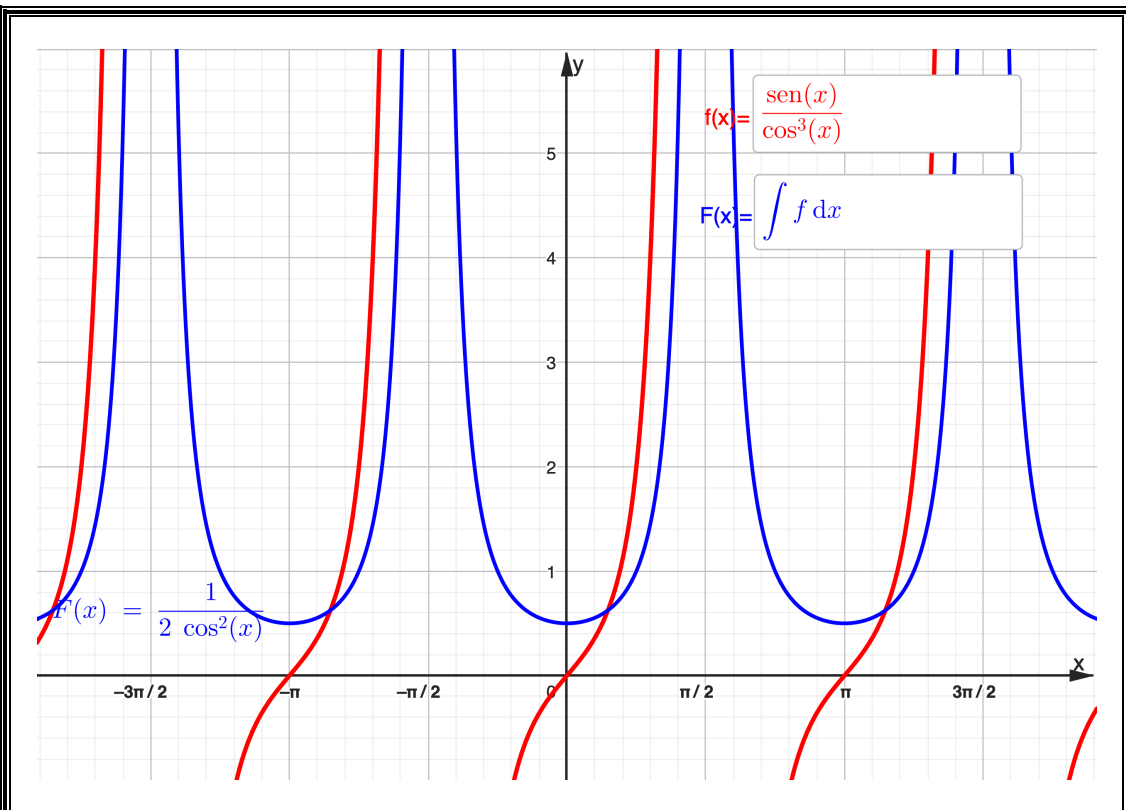
$$= - \int \frac{1}{u^3} du$$

$$= - \int u^{-3} du$$

$$= - \left(-\frac{u^{-2}}{2} \right) + C$$

$$= \frac{1}{2u^2} + C$$

$$= \frac{1}{2\cos^2(\beta)} + C$$



Ejemplo: Evaluar

$$\int \sqrt{1 + \cos(2x)} \, dx$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

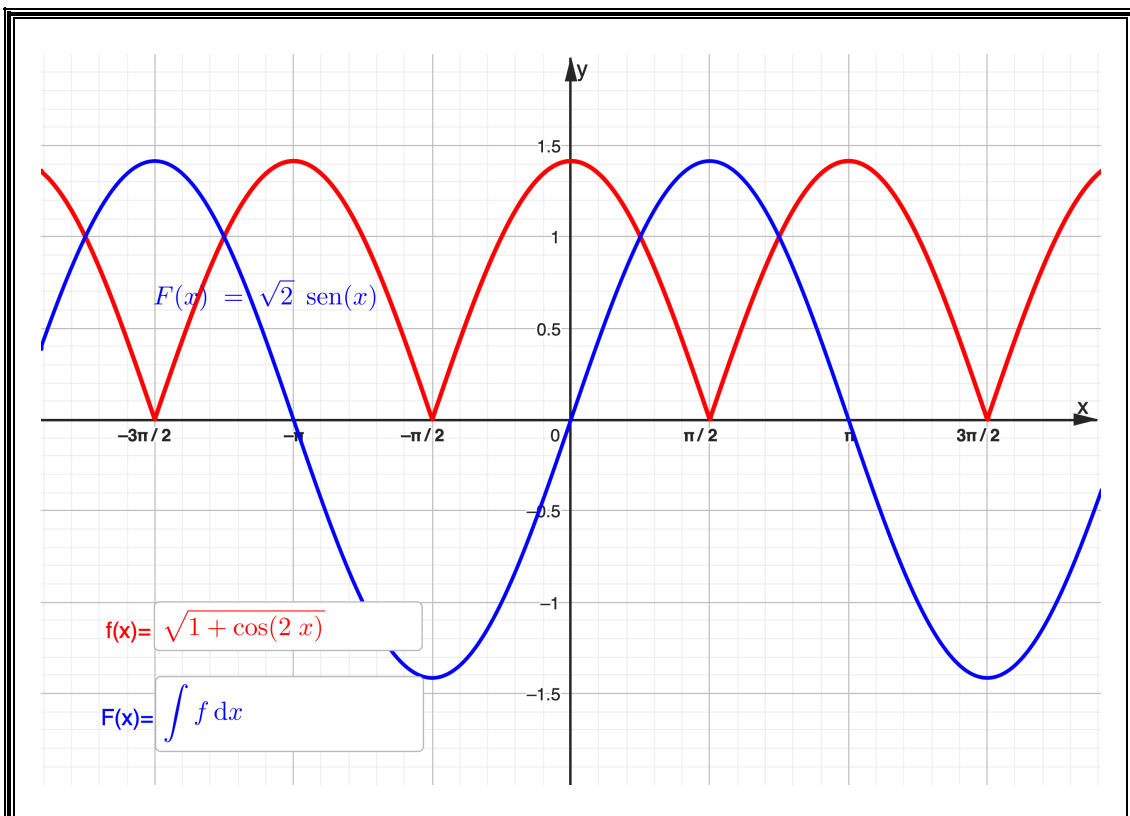
$$2\cos^2(x) = 1 + \cos(2x)$$

$$\sqrt{2} \cos(x) = \sqrt{1 + \cos(2x)}$$

$$\int \sqrt{2} \cos(x) \, dx$$

$$= \sqrt{2} \int \cos(x) \, dx$$

$$= \sqrt{2} \sin(x) + C$$



REDUCTION FORMULAS FOR SINE AND COSINE

The following reduction formulas can be used to integrate $\sin^k(x)$ and $\cos^k(x)$ for any exponent k , even or odd.

$$\int \sin^k(x) \, dx = -\frac{1}{k} \sin^{k-1}(x) \cos(x) + \frac{k-1}{k} \int \sin^{k-2}(x) \, dx$$

$$\int \cos^k(x) \, dx = \frac{1}{k} \cos^{k-1}(x) \sin(x) + \frac{k-1}{k} \int \cos^{k-2}(x) \, dx$$

Ejercicio: Evaluate $\int \sin^4(x) \, dx$ using the reduction formula

$$\int \tan^k(x) \, dx = \frac{\tan^{k-1}(x)}{k-1} - \int \tan^{k-2}(x) \, dx$$

$$\int \sec^k(x) \, dx = \frac{\tan(x) \sec^{k-2}(x)}{k-1} + \frac{k-2}{k-1} \int \sec^{k-2}(x) \, dx$$

EJERCICIOS

Evaluar las siguientes integrales y verificar la solución derivando:

a.	$\int \sin^2(x) \cos^3(x) dx$
b.	$\int \sin^2(x) \cos^2(x) dx$
c.	$\int \tan^3(x) dx$
d.	$\int \tan^3(x) \sec^4(x) dx$
e.	$\int \sin^3(x) dx$
f.	$\int \sin^2(x) \cos(x) dx$
g.	$\int \tan(x) \sec^2(x) dx$
h.	$\int \frac{\tan(\ln(x))}{x} dx$

RESPUESTAS DE LOS EJERCICIOS

a.	$\frac{\sin^3(x)}{3} - \frac{\sin^5(x)}{5} + C$
b.	$\frac{1}{8}x - \frac{\sin(4x)}{32} + C$
c.	$\frac{1}{2 \cos^2(x)} + \ln(\cos(x)) + C$
d.	$\frac{\sec^6(x)}{6} - \frac{\sec^4(x)}{4} + C$
e.	$\frac{\cos^3(x)}{3} - \cos(x) + C$
f.	$-\frac{\sin^3(x)}{3} + C$
g.	$\frac{\tan^2(x)}{2} + C$
h.	$\ln \sec(\ln(x)) + C$

